

# Product Catalog

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July 2013 - December 2013



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## Welcome

Maxeler provides a wide range of hardware and software products to allow you to exploit dataflow computing technology to address your computational and business challenges. If you know precisely which Maxeler product you would like information on, please turn to the page for that specific product.

If you would like to specify a complete MaxRack high performance computing system, we recommend starting on [page 4](#)

If you would like to specify a network processing system, we recommend starting on [page 20](#)





# Maxeler Product Range



## Dataflow Engines shared over Infiniband

MPC-X series dataflow nodes provide multiple dataflow engines as shared resources on the network, allowing them to be used by applications running anywhere on the cluster. The MPC-X series provides maximum flexibility and performance density for the widest range of HPC applications.

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## Tightly integrated Dataflow Engines and CPUs

MPC-C series nodes have the closest coupling of dataflow engines and multi-core CPUs within the same chassis. The MPC-C series are suitable for small systems or HPC applications requiring the most data transfer between CPUs and DFEs.

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## Low latency connectivity

MPC-N series nodes provide direct network connections to dataflow engines allowing ultra low latency line-rate processing of multiple 10-40Gbit data streams.

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## Dataflow computing on the desktop

MaxWorkstation and MaxWorkstation10G are small form factor PCs that bring the power of dataflow computing to the desktop. Ideal for software developers as well as users who desire a dedicated compute resource on their desk.

[Page 26](#)

## Multi-core CPU servers

1U, 2U or 1/2 U multi-core compute nodes, ready to connect to MPC-X series dataflow nodes or act as head or management nodes in a MaxRack cluster.

[Page 10](#)

## High Performance storage

Rack-level storage systems to add shared storage capacity to a MaxRack cluster.

[Page 14](#)

## MaxRack infrastructure

12U 'in-office', 42U or 48U data center racks with network switches and power distribution.

[Page 14](#)

## Development tools

General purpose development tools for Maxeler systems as well as domain specific languages to enable rapid development of applications in specific problem domains.

[Page 28](#)

## Securities trading interfaces

Plugins to provide full support for market data and order management for security trading.

[Page 29](#)

## Dataflow Engines

Specifications for each architecture of Maxeler Dataflow Engine (DFE)

[Page 30](#)

# Choosing a Maxeler system for HPC

## 1. Choose a system arrangement that matches your requirements

Page 6

HPC clusters with dataflow computing technology can be built around different combinations of MPC-X series and MPC-C series dataflow nodes.

- MPC-X series systems provide both high density and flexibility. MPC-X nodes can be connected directly to CPU-only nodes via single or dual Infiniband cables, or via an Infiniband switch to share dataflow engines across the network.
- MPC-C series systems are ideal for small deployments, or for applications that require the most data transfer between DFEs and CPUs.

## 2. Configure your MPC nodes and CPU-only nodes

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Maxeler MPC nodes and CPU only nodes can be configured to match a wide range of application requirements.

- MPC-X series nodes can be configured with the type and number of DFEs, amount of DFE RAM & interconnect. [Page 7](#)
- MPC-C series nodes can be configured with the type and number of DFEs, amount of DFE RAM, CPU type and main memory size, in-node disk storage and interconnect. [Page 8](#)
- CPU only nodes can be configured to select CPU type and main memory size, storage and interconnect. [Page 10](#)

## 3. Do you want separate "Cluster Management" node(s)?

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"Cluster Management" nodes are standard CPU nodes which run software to perform functions such as job distribution, database hosting or system provisioning. If a MaxRack system is intended to function standalone, management nodes are recommended. If you intend to integrate into a larger cluster, management nodes may not be necessary. Management nodes need not have the same configuration as other CPU nodes in a MaxRack cluster.

- For system management MaxRack head nodes run xcat cluster management software.
- For job management, MaxRack head nodes can run open source job distribution systems such as Grid Engine or Maxeler can develop a custom engineered solution for your requirements using Maxeler ClusterMap technology.

## 4. Choose network architecture & network switches

Page 12

You can choose a network architecture depending on the system arrangement you have selected and how you wish to integrate into the rest of the data center. Typically MaxRack systems contain a low-speed Ethernet management network (used to remotely monitor and control nodes) and a high-speed data network utilizing Ethernet and/or Infiniband. Options include:

- Single 1Gb Ethernet network for data and management. Requires at least 1 Ethernet switch.
- 1Gb Ethernet management network, Infiniband data network. Requires at least 1 Ethernet and 1 Infiniband switch.
- 1Gb Ethernet management network, 1GbE or 10GbE data network. Requires at least 2 Ethernet switches.
- 1Gb Ethernet management network, Infiniband data network and Ethernet data network. Requires at least 2 Ethernet switches and 1 Infiniband switch.

## 5. Do you want dedicated shared storage for the cluster?

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A MaxRack cluster can make use of dedicated shared storage to hold input/output data or shared scratch data for easy access by the cluster nodes. Rack-level storage can be optimized for the needs of the MaxRack system.

## 6. Choose power feed type

Page 14

It is important to specify power feeds that match the types available in your deployment environment and your requirements for the MaxRack system. Power distribution units (PDUs) in the MaxRack will distribute power from the input feed to individual nodes.

- MaxRack systems can be configured with 16A, 32A or 3-phase power feeds. For office environments, small MaxRack systems can be connected directly to multiple wall sockets.
- For power feed redundancy, make sure to only utilize nodes that support redundant power supplies then specify a complete set of PDUs for each feed. Each node should be connected to one PDU on feed A and one PDU on feed B, and PDUs should only be loaded to half their rated current to allow them to continue functioning if one feed fails.

## 7. Choose enclosure form factor

Page 15

Rack enclosures must have enough space to support all the nodes, switches and power distribution you have selected. Multiple racks may be necessary for large systems.

- For data center environments, we recommend either 42U or 48U rack enclosures
- For office environments, we can provide 12U enclosed racks with sound proofing and active noise-cancelling.

## 8. Check overall requirements against data center requirements

Once you've configured your MaxRack system, you can double check the system against the requirements for your data center. We recommend checking the following information in particular:

- ☐ Rack enclosure size support
- ☐ Size and number of racks to support required number of nodes, switches, PDUs and cabling
- ☐ Type and number of external network connections
- ☐ Type and number of external power feeds
- ☐ Maximum power dissipation per rack supported by data center cooling
- ☐ Floor loading requirement

## 9. Choose development software

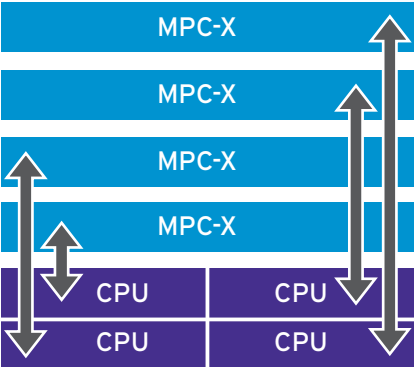
Page 28

MaxCompiler is required for programming Maxeler dataflow computing systems. Make sure to select the MaxCompiler edition supporting the DFE architecture in your MPC nodes. In addition, you may wish to utilize our domain specific languages which build on top of MaxCompiler to add particular support for specific application areas:

- For 3D finite difference applications such as seismic Forward Modeling, Reverse Time Migration and Full Waveform Inversion, we recommend MaxGenFD.



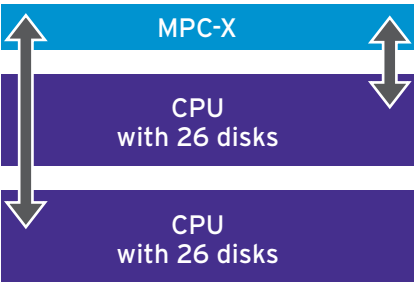
# Common compute system arrangements



## Directly connected for compute bound applications

MPC-X series nodes can be directly connected to CPU nodes via Infiniband cables without a switch. Each MPC-X node can have 1 or 2 Infiniband connections which can be connected to a single CPU node (to provide maximum bandwidth between the MPC-X DFEs and the CPU node) or two CPU nodes (to allow both CPU nodes to share a single MPC-X node).

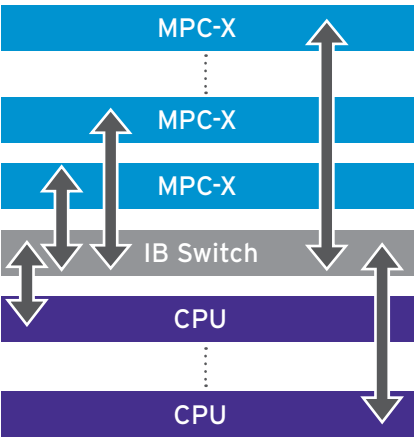
In the illustrated configuration, MPC-X nodes are connected to high density CPU-C nodes, providing 32 DFEs and 4 CPU nodes in 6U, with a ratio of 8 DFEs to 1 CPU node. Each CPU node can utilize 1 or 2 Infiniband uplinks depending on the desired bandwidth.



## Directly connected for disk bound applications

For disk bound applications, DFE compute resource and disk capability must be balanced. If the application requirements are static, MPC-X series nodes can be directly connected to high disk capability CPU-B nodes.

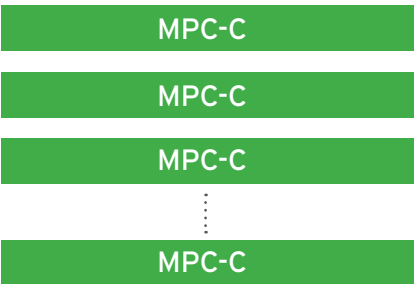
In the illustrated configuration, a single MPC-X node is directly connected to two CPU-B nodes, providing 8 DFEs and 2 CPU nodes in 5U with a ratio of 4 DFEs to 1 CPU node.



## Switched DFE-CPU allocation for dynamic applications

Where application requirements are not static at runtime, or multiple applications with differing requirements should be able to run on the same hardware, MPC-X nodes provide the most flexibility by supporting a switched interconnection between multiple MPC-X nodes and multiple CPU nodes. Any CPU node can utilize any number of DFEs on the Infiniband network and utilization of DFEs is maximized by making them available to any CPU node that might wish to use them. Excess CPU node capacity can be utilized to run CPU-only processes.

In the illustrated configuration, multiple MPC-X series nodes and multiple general-purpose CPU-A nodes are connected to a 1U Infiniband switch.



## Homogenous cluster of nodes

For smaller systems, a homogeneous cluster of MPC-C series nodes can be the most suitable solution. Individual MPC-C nodes can be configured to handle the demands of the application and then this single node unit can be replicated to build larger systems.

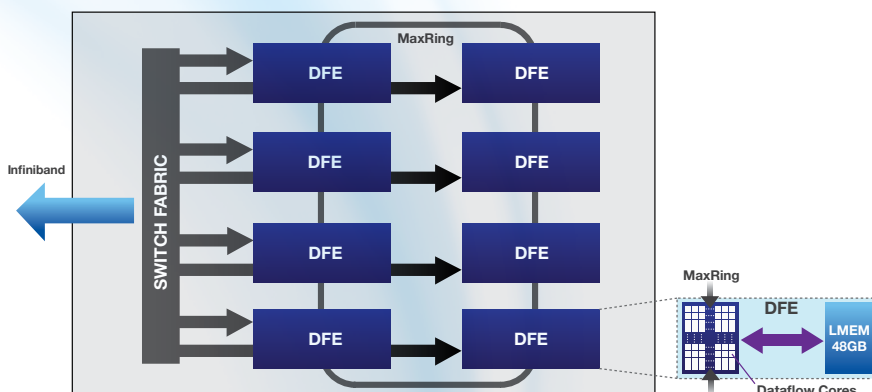
MPC-C series clusters are also well suited for applications which demand the highest levels of bandwidth between DFEs and CPUs, with a direct PCI Express connection to each DFE.



# MPC-X Series

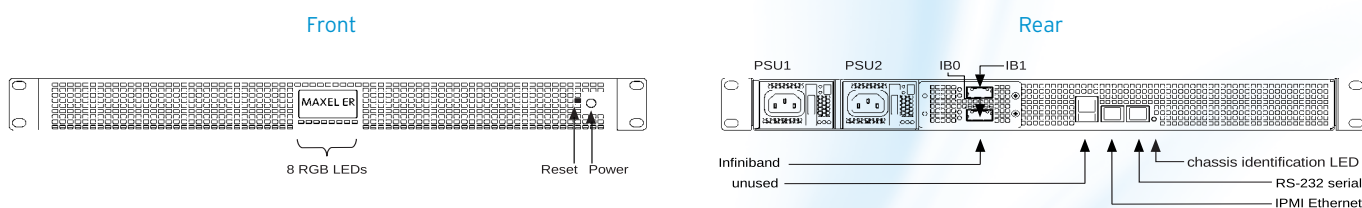


MPC-X series dataflow nodes provide multiple dataflow engines as shared resources on the network, allowing them to be used by applications running anywhere on the cluster. The MPC-X series provides maximum flexibility and performance density for the widest range of HPC applications.



MPC-X Series Architecture

## MPC-X Chassis



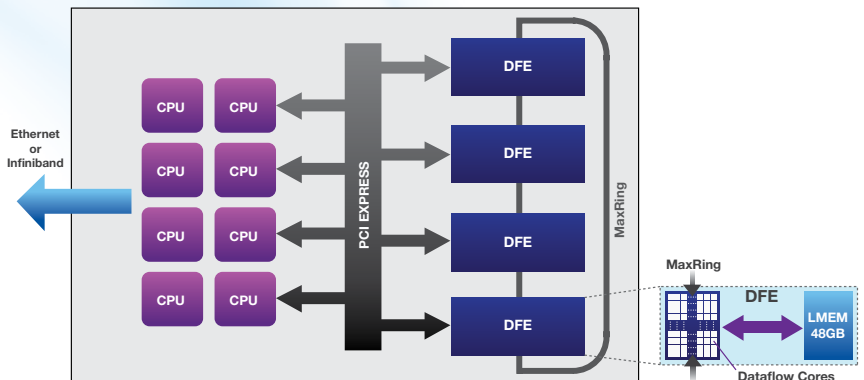
## Specifications

|                          | MPC-X1000                            | MPC-X1200                            | MPC-X2000                            |
|--------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| Form factor              | 1U                                   |                                      |                                      |
| DFE Architecture         | <i>vectis</i>                        | <i>coria</i>                         | <i>maia</i>                          |
| DFE LMEM                 | 24 GB per DFE                        | 24 GB per DFE<br>or<br>48 GB per DFE | 24 GB per DFE<br>or<br>48 GB per DFE |
| Number of DFEs           | 1 or 4 or 8                          |                                      |                                      |
| MaxRing                  | ✓ <sup>†</sup>                       | ✓                                    | ✓                                    |
| Interconnect             | 1x Infiniband<br>or<br>2x Infiniband |                                      |                                      |
| Redundant power supplies | ✓                                    |                                      |                                      |
| Power consumption        | 830W                                 | 850W                                 | 950W                                 |
| Weight                   | 21kg                                 |                                      |                                      |
| Dimensions               | 434mm x 43mm x 840mm                 |                                      |                                      |
| Availability             | Now                                  | Now                                  | Now                                  |

<sup>†</sup> 2 independent 1D networks of 4 DFEs

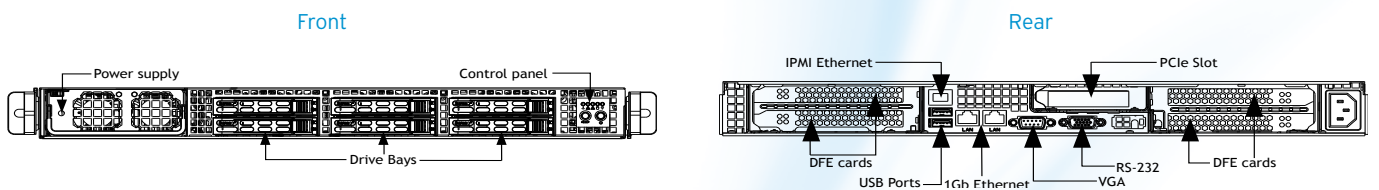
# MPC-C Series

MPC-C series nodes have the closest coupling of dataflow engines and multi-core CPUs within the same chassis. The MPC-C series are suitable for small systems or HPC applications requiring the most data transfer between CPUs and DFEs.

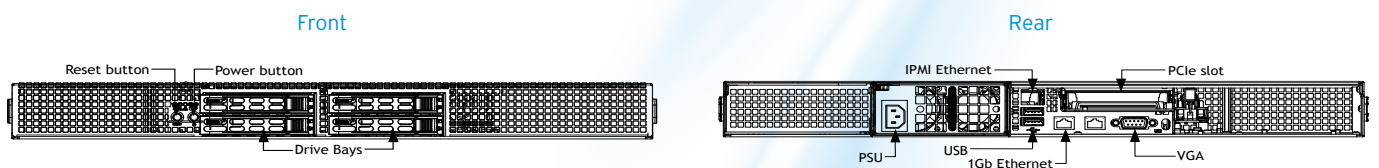


MPC-C Series Architecture

## MPC-C500 Chassis



## MPC-C600/1000 Chassis





## Specifications

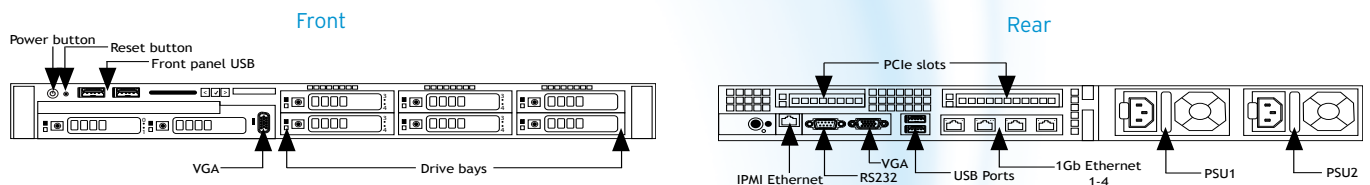
|                          | MPC-C500                                                                                                                                             | MPC-C600                                                                                       | MPC-C1000   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------|
| Form factor              | 1U                                                                                                                                                   |                                                                                                |             |
| DFE Architecture         | <i>vectis</i>                                                                                                                                        | <i>coria</i>                                                                                   | <i>maia</i> |
| DFE LMEM                 | 24 GB per DFE                                                                                                                                        | 24 GB per DFE<br>or<br>48 GB per DFE                                                           |             |
| Number of DFEs           | 1, 2, 3 or 4                                                                                                                                         |                                                                                                |             |
| CPU                      | 2x Intel Xeon X5650<br>12 cores @ 2.66GHz<br>or<br>2x Intel Xeon X5680<br>12 cores @ 3.33GHz                                                         | 2x Intel Xeon E5-2640<br>12 cores @ 2.5GHz<br>or<br>2x Intel Xeon E5-2670<br>16 cores @ 2.6GHz |             |
| Memory                   | 48GB<br>or<br>96GB<br>or<br>192GB                                                                                                                    | 64GB<br>or<br>128GB                                                                            |             |
| Interconnect             | 2x 1GbE<br>or<br>2x 1GbE and 1x FDR Infiniband<br>or<br>2x 1GbE and 2x 10GbE                                                                         |                                                                                                |             |
| Storage                  | 2x1TB in RAID1 (1TB usable)<br>or<br>4x1TB in RAID5 (3TB usable)<br>or<br>2x1TB in RAID1 + 2x 240GB SSD in RAID0 (1TB system + 480GB scratch usable) |                                                                                                |             |
| RAID                     | Software                                                                                                                                             |                                                                                                |             |
| Operating System         | CentOS 5 / RHEL 5<br>or<br>CentOS 6 / RHEL 6                                                                                                         | CentOS 6 / RHEL 6                                                                              |             |
| MaxRing                  | ✓                                                                                                                                                    |                                                                                                |             |
| Redundant power supplies |                                                                                                                                                      |                                                                                                |             |
| Power consumption        | 560W (X5650 CPUs)                                                                                                                                    | 600W (E5-2640 CPUs)                                                                            |             |
| Dimensions               | 437mm x 43mm x 716mm                                                                                                                                 | 437mm x 43mm x 876mm                                                                           |             |
| Weight                   | 17kg                                                                                                                                                 | 18kg                                                                                           | 18kg        |
| Availability             | Now                                                                                                                                                  | Q4 2013                                                                                        | Now         |

# CPU Servers

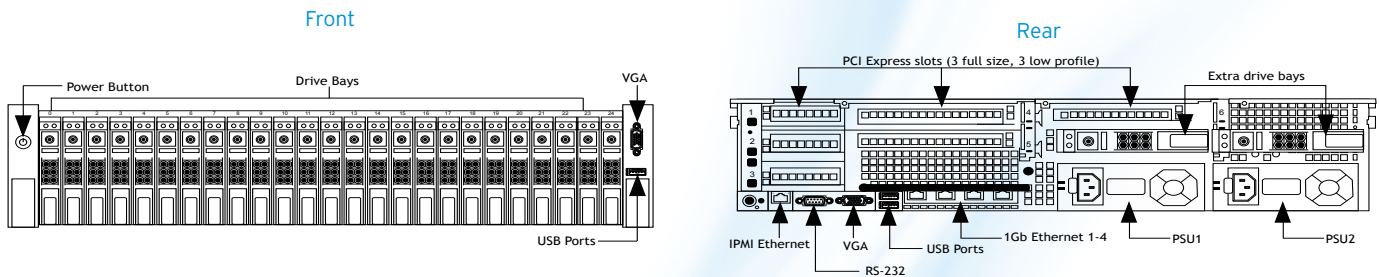
Maxeler can offer a range of standard HPC servers to accompany our MPC-X Series compute nodes or otherwise form part of a MaxRack system.

Select CPU-A for a balanced system configuration, CPU-B for maximum disk performance or CPU-C for maximum compute density.

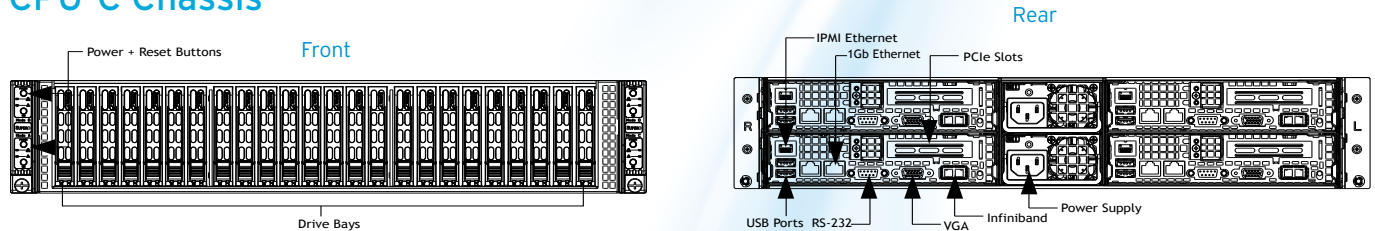
## CPU-A Chassis



## CPU-B Chassis



## CPU-C Chassis



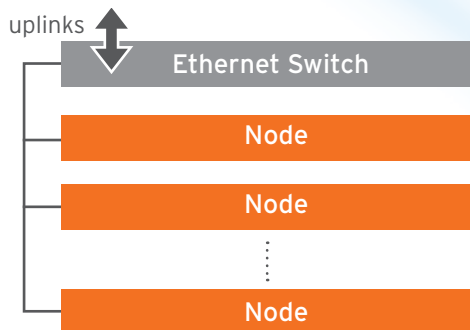
\* Please note that the labels are applicable for the other 3 nodes in the chassis

## Specifications

|                          | CPU-A                                                                                                                                                                          | CPU-B                                                                                                                                                                                 | CPU-C                                                                                                                                                                           |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form factor              | 1U                                                                                                                                                                             | 2U                                                                                                                                                                                    | 4 nodes in 2U                                                                                                                                                                   |
| CPU                      | 2x Intel Xeon E5-2640<br>12 cores @ 2.5GHz<br><b>or</b><br>2x Intel Xeon E5-2670<br>16 cores @ 2.6GHz<br><b>or</b><br>2x Intel Xeon E5-2690<br>16 cores @ 2.9GHz               |                                                                                                                                                                                       | 2x Intel Xeon E5-2640<br>12 cores @ 2.5GHz<br><b>or</b><br>2x Intel Xeon E5-2670<br>16 cores @ 2.6GHz                                                                           |
| Memory                   | 64GB<br><b>or</b><br>192GB<br><b>or</b><br>384GB                                                                                                                               |                                                                                                                                                                                       | 64GB<br><b>or</b><br>128GB                                                                                                                                                      |
| Interconnect             | 4x 1GbE and 1x FDR IB<br><b>or</b><br>4x 1GbE and 2x FDR IB<br><b>or</b><br>4x 1GbE and 2x 10GbE<br><b>or</b><br>4x 1GbE and 1x FDR IB and 2x 10GbE                            |                                                                                                                                                                                       | 2x 1GbE and 1x FDR IB<br><b>or</b><br>2x 1GbE and 2x FDR IB<br><b>or</b><br>2x 1GbE and 1x FDR IB<br>and 2x 10GbE                                                               |
| Storage                  | 2x1TB in RAID1<br>(1TB usable)<br><b>or</b><br>4x1TB in RAID5<br>(3TB usable)<br><b>or</b><br>2x1TB in RAID1 + 2x<br>240GB SSD in RAID0<br>(1TB O/S + 480GB<br>scratch usable) | 2x1TB in RAID1 for<br>system plus 24x data<br>disks in 2 RAID5 groups:<br><br>1TB (22TB usable)<br><b>or</b><br>600GB 10k RPM<br>(13TB usable)<br><b>or</b><br>240GB SSD (5TB usable) | 2x1TB in RAID1<br>(1TB usable)<br><b>or</b><br>4x1TB in RAID5<br>(3TB usable)<br><b>or</b><br>2x1TB in RAID1 + 2<br>x 240GB SSD in RAID0<br>(1TB O/S + 480GB<br>scratch usable) |
| RAID                     | Hardware                                                                                                                                                                       | Hardware                                                                                                                                                                              | Software                                                                                                                                                                        |
| Operating System         | CentOS 6 / RHEL 6                                                                                                                                                              |                                                                                                                                                                                       |                                                                                                                                                                                 |
| Redundant power supplies | ✓                                                                                                                                                                              | ✓                                                                                                                                                                                     | ✓                                                                                                                                                                               |
| Power consumption        | 400W (E5-2670 CPUs)                                                                                                                                                            | 450W (E5-2670 CPUs,<br>24 data disks)                                                                                                                                                 | 400W (E5-2670 CPUs)<br>per node                                                                                                                                                 |
| Dimensions               | 437mm x 43mm x<br>736mm                                                                                                                                                        | 444mm x 88mm x<br>723mm                                                                                                                                                               | 438mm x 88mm x<br>697mm                                                                                                                                                         |
| Weight                   | 19kg                                                                                                                                                                           | 33kg (24 data disks)                                                                                                                                                                  | 39kg per 4 nodes                                                                                                                                                                |
| Availability             | Now                                                                                                                                                                            | Now                                                                                                                                                                                   | Now                                                                                                                                                                             |

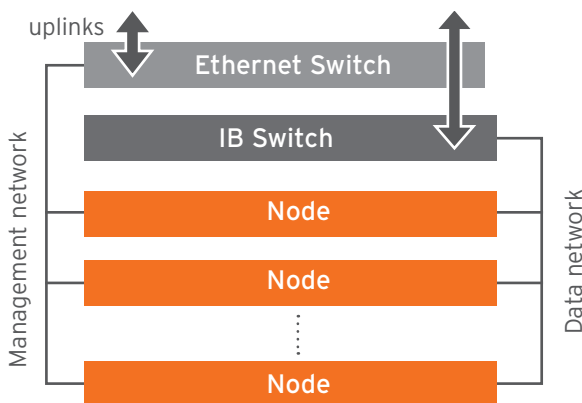


# Network Architectures



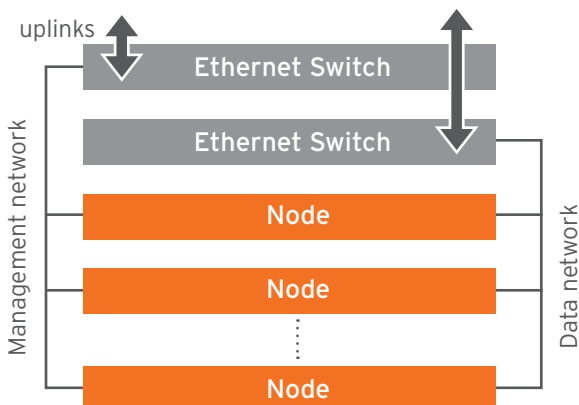
## Single 1Gb Ethernet network for data and management

Ideal for small systems with a few nodes and low data movement requirements, requiring only a single Ethernet switch.



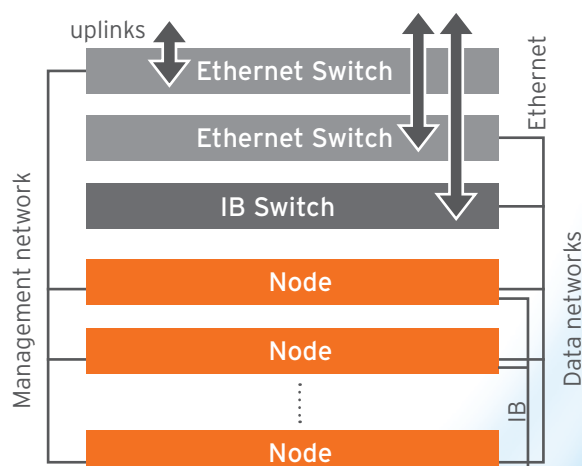
## 1Gb Ethernet management network, Infiniband data network

Suitable for MPC-X Series based clusters with switched interconnect between MPC-X nodes and CPU nodes, or clusters with high data transfer requirements. If the MaxRack should be connected to an external data network based on Ethernet, a CPU head node equipped with both Infiniband and Ethernet interfaces can be used to bridge between the cluster's Infiniband network and the external Ethernet network. Requires at least 1 Ethernet and 1 Infiniband switch.



## 1Gb Ethernet management network, 1GbE or 10GbE data network

Suitable where compute nodes need individual connections to an upstream Ethernet data network.



## 1Gb Ethernet management network, Infiniband data network and Ethernet data network

Suitable where the MaxRack requires a high-bandwidth internal network between nodes, or MPC-X series nodes are connected to CPU nodes via a switched interconnect, but each node also requires a direct connection to an external Ethernet data network. The Ethernet data network could be either 1Gb or 10Gb.

# Network Switches

## Ethernet Switches

|                          | Cisco Nexus 5548UP | Cisco Nexus 3048         |
|--------------------------|--------------------|--------------------------|
| Form factor              | 1U                 |                          |
| Ports                    | 48 x 10GbE         | 48 x 1GbE + 4x10GbE SFP+ |
| Redundant power supplies | ✓                  | ✓                        |
| Power consumption        | 390W               | 120W                     |
| Weight                   | 16kg               | 9.3kg                    |

## Infiniband Switches

|                          | Mellanox SX6506 | Mellanox SX6025 |
|--------------------------|-----------------|-----------------|
| Form factor              | 7U              | 1U              |
| Ports                    | 108             | 36              |
| Redundant power supplies | ✓               | ✓               |
| Power consumption        | 1400W           | 113W            |
| Weight                   | 80kg            | 9.3kg           |

# MaxRack Infrastructure

## Power distribution

The choice of specific power distribution units for your Maxeler system depends on type of power feeds you wish to use (voltage, current, 1-phase/3-phase and connector type). For rack planning purposes we recommend using the details below.

|               | 16A Feed                       | 32A Feed                       | 3 Phase                                  |
|---------------|--------------------------------|--------------------------------|------------------------------------------|
| Form factor   | 1U                             | 1U                             | Vertical (2 fit per rack)                |
| Input current | 16A                            | 32A                            | 3 phase                                  |
| Num outputs   | 8                              | 8                              | 24                                       |
| Maximum load  | 3,680W @ 230V<br>3,328W @ 208V | 7,360W @ 230V<br>6,656W @ 208V | 22,080W @ 32A-230V<br>19,968W @ 32A-208V |

## Racks

|                    | 12U Enclosed<br>'Quiet rack' | 42U data centre            | 48U data centre            |
|--------------------|------------------------------|----------------------------|----------------------------|
| Rack units         | 12U                          | 42U                        | 48U                        |
| Dimensions (WxDxH) | 716mm x 780mm x<br>1210mm    | 600mm x 1200mm x<br>1991mm | 750mm x 1200mm x<br>2258mm |
| Weight             | 118kg                        | 134kg                      | 186kg                      |

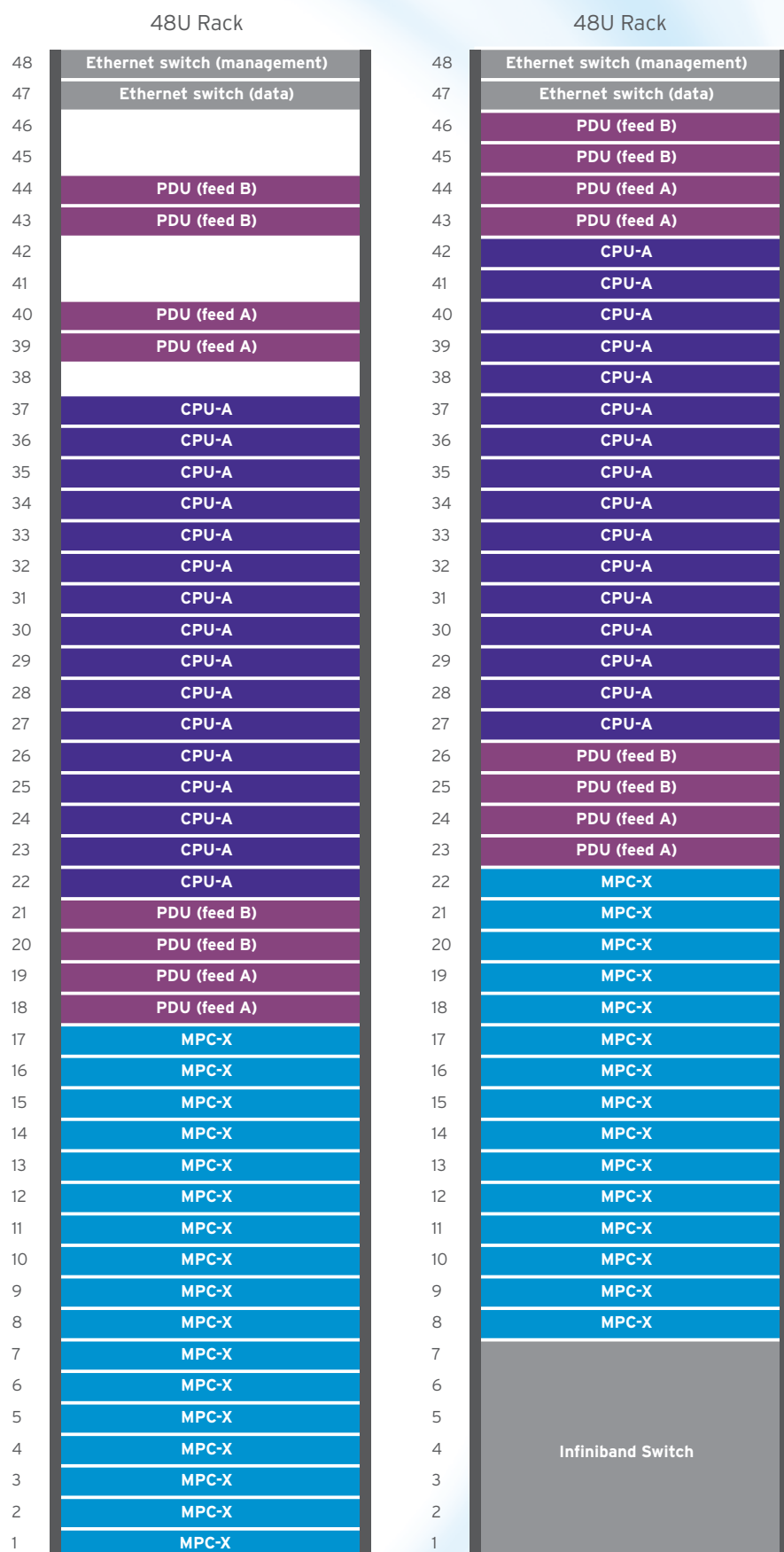
## High Performance Storage

You may choose to add additional shared cluster-level storage to your MaxRack. This can be beneficial as a staging area for input data, temporary store for scratch data, or to consolidate output data. A shared fileserver exporting NFS to client machines offers a good balance for most applications or for ultimate performance scalability, you can specify a Lustre-based parallel file system.

|                          | NFS Fileserver                                                            | Lustre Parallel File System                                                      |
|--------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Form factor              | 2U                                                                        | 2U + 2U per unit (minimum 2 units, 6U)                                           |
| Interconnect             | 4x 1GbE and 1x FDR IB<br>or<br>4x 1GbE and 2x10GbE                        | 1x FDR Infiniband + 1x FDR<br>Infiniband per unit                                |
| Main storage             | 20TB for linear I/O (2x RAID5,10+1)<br>or<br>11TB for random I/O (RAID10) | Per unit<br>11TB for random I/O (RAID10)                                         |
| Hot spare drives         | 2                                                                         | 2 per unit                                                                       |
| Redundant power supplies | ✓                                                                         | ✓                                                                                |
| Redundant servers        |                                                                           | Optional<br>(add additional 2U head node and<br>specify double storage capacity) |
| Power consumption        | 450W                                                                      | 530W + 450W per storage unit                                                     |
| Dimensions               | 444mm x 88mm x 723mm                                                      | 444mm x 88mm x 723mm<br>for head node and for each storage unit                  |
| Weight                   | 33kg                                                                      | 29kg + 33kg per unit (minimum 111kg)                                             |
| Availability             | Now                                                                       | Now                                                                              |



# Example MaxRack Systems



## Analytics Cluster

This configuration is optimized for dynamic applications consisting of many small processing jobs, such as financial risk analytics. Any CPU in the system can utilize any DFE, allowing optimum load balancing across the cluster.

### Key statistics:

- Two 48U racks
- 256 DFEs in 32 MPC-X nodes
- 512 CPU cores in 32 CPU nodes
- 8 DFEs per CPU node
- 42kW power consumption
- Redundant 32A power feeds

# Example MaxRack Systems

48U Rack

|    |                              |
|----|------------------------------|
| 48 | Ethernet switch (management) |
| 47 | Ethernet switch (data)       |
| 46 |                              |
| 45 | PDU                          |
| 44 | PDU                          |
| 43 | PDU                          |
| 42 | PDU                          |
| 41 |                              |
| 40 | MPC-X                        |
| 39 | CPU-B                        |
| 38 |                              |
| 37 | CPU-B                        |
| 36 |                              |
| 35 | MPC-X                        |
| 34 | CPU-B                        |
| 33 |                              |
| 32 | CPU-B                        |
| 31 |                              |
| 30 | MPC-X                        |
| 29 | CPU-B                        |
| 28 |                              |
| 27 | CPU-B                        |
| 26 |                              |
| 25 | MPC-X                        |
| 24 | CPU-B                        |
| 23 |                              |
| 22 | CPU-B                        |
| 21 |                              |
| 20 | MPC-X                        |
| 19 | CPU-B                        |
| 18 |                              |
| 17 | CPU-B                        |
| 16 |                              |
| 15 | MPC-X                        |
| 14 | CPU-B                        |
| 13 |                              |
| 12 | CPU-B                        |
| 11 |                              |
| 10 | MPC-X                        |
| 9  | CPU-B                        |
| 8  |                              |
| 7  | CPU-B                        |
| 6  |                              |
| 5  | MPC-X                        |
| 4  | CPU-B                        |
| 3  |                              |
| 2  | CPU-B                        |
| 1  |                              |

## Cluster for I/O intensive applications

This configuration is optimized for I/O intensive applications, such as seismic Reverse Time Migration with stored snapshots. The cluster balances DFE compute resource with high disk capacity and performance.

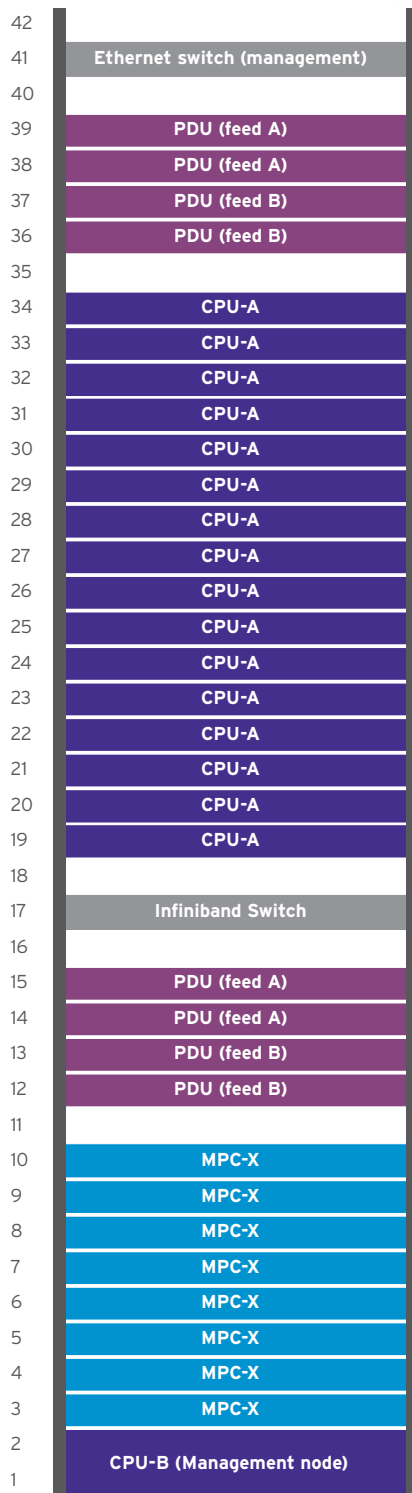
The cluster is built from a repeating unit of a single MPC-X series dataflow node connected to two disk-intensive CPU-B nodes. Each CPU-B node uses 24 high performance disks to store temporary data generated during the computation. MPC-X nodes are statically connected to the CPU nodes via direct Infiniband cables, so no Infiniband switches are required.

### Key statistics:

- One 48U rack
- 64 DFEs in 8 MPC-X nodes
- 256 CPU cores in 16 CPU nodes
- 4 DFEs per CPU node
- 14kW power consumption
- 32A power feeds (non-redundant)



42U Rack



## Mid-sized Cluster for compute intensive applications

This configuration is optimized for medium-scale compute intensive applications. The compute cluster is self-contained, with a CPU management node providing job distribution, management and results aggregation.

The cluster uses CPU-A nodes which can utilize DFEs from multiple MPC-X nodes. The main data network is provided with a single 36-port Infiniband switch of which 3A ports are utilized for the CPU nodes (1 port per node), MPC-X nodes (2 ports per node) and management node (2 ports). Multiple clusters of this size can be connected together using the remaining 2 ports as uplinks between the two cluster switches.

### Key statistics:

- One 42U rack
- 64 DFEs in 8 MPC-X nodes
- 256 CPU cores in 16 CPU nodes
- 4 DFEs per CPU node
- 15kW power consumption
- Redundant 32A power feeds

# Example MaxRack Systems

42U Rack

|    |                              |     |
|----|------------------------------|-----|
| 42 | CPU-B (Management node)      |     |
| 41 |                              |     |
| 40 | Ethernet switch (management) |     |
| 39 | PDU                          |     |
| 38 | PDU                          |     |
| 37 | CPU                          | CPU |
| 36 | CPU                          | CPU |
| 35 | CPU                          | CPU |
| 34 | CPU                          | CPU |
| 33 | CPU                          | CPU |
| 32 | CPU                          | CPU |
| 31 | Infiniband Switch            |     |
| 30 | MPC-X                        |     |
| 29 | MPC-X                        |     |
| 28 | MPC-X                        |     |
| 27 | MPC-X                        |     |
| 26 | MPC-X                        |     |
| 25 | MPC-X                        |     |
| 24 | MPC-X                        |     |
| 23 | MPC-X                        |     |
| 22 | MPC-X                        |     |
| 21 | MPC-X                        |     |
| 20 | PDU                          |     |
| 19 | PDU                          |     |
| 18 | PDU                          |     |
| 17 | CPU                          | CPU |
| 16 | CPU                          | CPU |
| 15 | CPU                          | CPU |
| 14 | CPU                          | CPU |
| 13 | CPU                          | CPU |
| 12 | CPU                          | CPU |
| 11 | Infiniband Switch            |     |
| 10 | MPC-X                        |     |
| 9  | MPC-X                        |     |
| 8  | MPC-X                        |     |
| 7  | MPC-X                        |     |
| 6  | MPC-X                        |     |
| 5  | MPC-X                        |     |
| 4  | MPC-X                        |     |
| 3  | MPC-X                        |     |
| 2  | MPC-X                        |     |
| 1  | MPC-X                        |     |

## High density compute cluster

This configuration is optimized for high compute density within a relatively small 42U data center rack. The cluster is self-contained with a CPU management node, and a large number of high density CPU-C nodes and MPC-X nodes providing compute resources.

The main data network is provided by two 36-port Infiniband switches. Each Infiniband switch has connections to 10 MPC-X nodes and 12 CPU nodes and also an uplink to the management node. The switches are cross-linked by up to 3 Infiniband connections. This provides a high density network with only 2U required for Infiniband switching, suiting applications where there is significant locality between a CPU node and the MPC-X node from which it will use DFEs, since there is somewhat limited bandwidth available between the two switches.

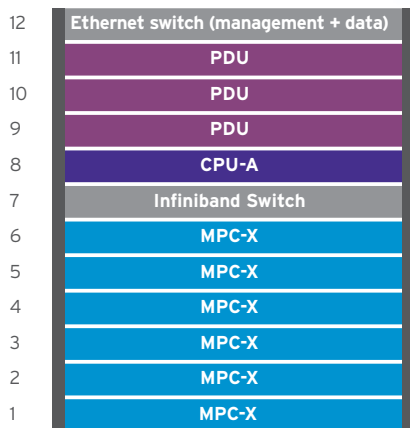
### Key statistics:

- One 42U rack
- 160 DFEs in 20 MPC-X nodes
- 384 CPU cores in 24 CPU nodes
- 6.7 DFEs per CPU node
- 30kW power consumption
- 32A power feeds (non-redundant)





12U Rack



## In-office Cluster for DFE-dominated computations

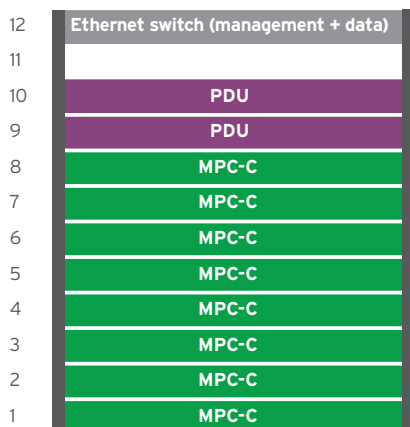
This configuration is for small-scale in-office deployments in a 12U sound-proof rack environment. The system is optimized for a high ratio of DFE computation to CPU computation, found in application such as cryptography, searching or pattern matching.

A single CPU node acts as the main node in the system, and dispatches work to multiple DFEs in MPC-X nodes.

### Key statistics:

- 12U sound-proof noise cancelling rack
- 48 DFEs in 6 MPC-X nodes
- 16 CPU cores in 1 CPU node
- 48 DFEs per CPU node
- 6kW power consumption
- Three 13A @ 230V wall socket power feeds

12U Rack



## In-office Cluster for balanced CPU/DFE computations

This configuration is for small-scale in-office deployments in a 12U sound-proof rack environment. The system uses MPC-C series nodes to provide a balanced (static) ratio of CPU cores to DFEs.

### Key statistics:

- 12U sound-proof noise cancelling rack
- 32 DFEs and 96 CPU cores in 8 MPC-C nodes
- 4 DFEs per CPU node
- 5kW power consumption
- Two 13A @ 230V wall socket power feeds

# Choosing a Maxeler system for networking

## 1. Select a system arrangement that matches your requirements

Page 21

Dataflow computing technology can be utilized at various points in a network, including as a client or server, data distributor, aggregator or gateway.

## 2. Configure MPC-N system(s)

Page 22

Each MPC-N series node can be configured with multiple network connected DFEs. Each isca DFE can support 1, 2 or 3 QSFP+ 40Gbit network connections (or each 40Gbit port can be split into 4x10Gbit port). vectis DFEs support 2x SFP+ 10Gbit network connections. All MPC-N Series nodes also support additional 40Gbit or 10Gbit connections direct to the CPUs and are available in either 1U or 2U form factors. You may select 1U form factors for maximum density, or 2U for expandability and resiliency with dual redundant power supplies and support for up to 16 hard drives.

- MPC-N320 is 1U form factor and provides up to 4 isca DFEs.
- MPC-N322 is 2U form factor with up to 4 isca DFEs.
- MPC-N40 is a 1U form factor system with up to 2 vectis DFEs.
- MPC-N42 is a 2U form factor system with up to 2 vectis DFEs.

## 3. Select network switch(es)

Page 13

For system arrangements that involve attaching MPC-N series systems to Ethernet switches, you can use any existing switch infrastructure, or acquire switches from Maxeler as part of your system.

## 4. Choose financial exchange interface libraries

Page 27

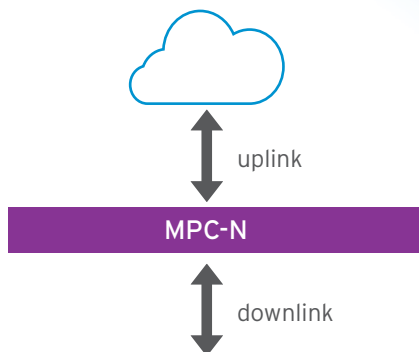
Maxeler's fully maintained Market Data and Order Management libraries offer seamless connectivity to financial venues. To connect to financial exchanges, select the interface libraries you would like to use.

## 5. Choose development tools

Page 26

MaxCompiler and MaxCompilerMPT are recommended for programming Maxeler networking systems.

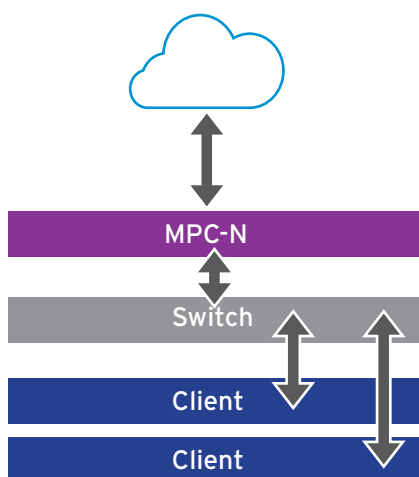
# Common networking **system arrangements**



## Gateway

MPC-N series nodes can function as gateways between different networks or as a bridge between one or multiple clients and upstream servers. Example application areas include firewalls, network forensics and Direct Market Access gateways.

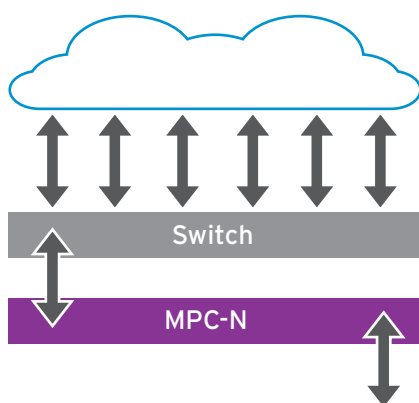
Each DFE in an MPC-N series node has two network ports, one of which can be connected to the upstream and one to the downstream network.



## Distribution system

Distribution arrangements pre-process data from an upstream network and then dispatch data to downstream clients. An example of this type of application is financial market data transformation, pre-processing exchange data feeds into a form optimized for use by downstream nodes.

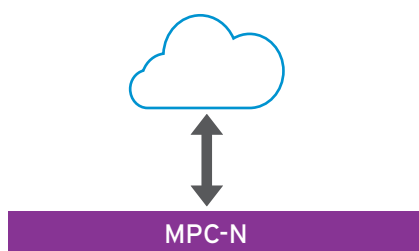
One network port in an MPC-N DFE can be connected to the upstream network and a second can be connected to a downstream Ethernet switch to distribute data to multiple downstream clients.



## Aggregator

MPC-N nodes can aggregate multiple upstream data feeds into a single feed, pre-processing and filtering data before passing a more manageable data stream to a downstream node.

One network port in an MPC-N DFE can be connected to an upstream network switch and the node can listen to data from many independent data sources connected to that switch. A second network port on the DFE can be connected to the downstream node.



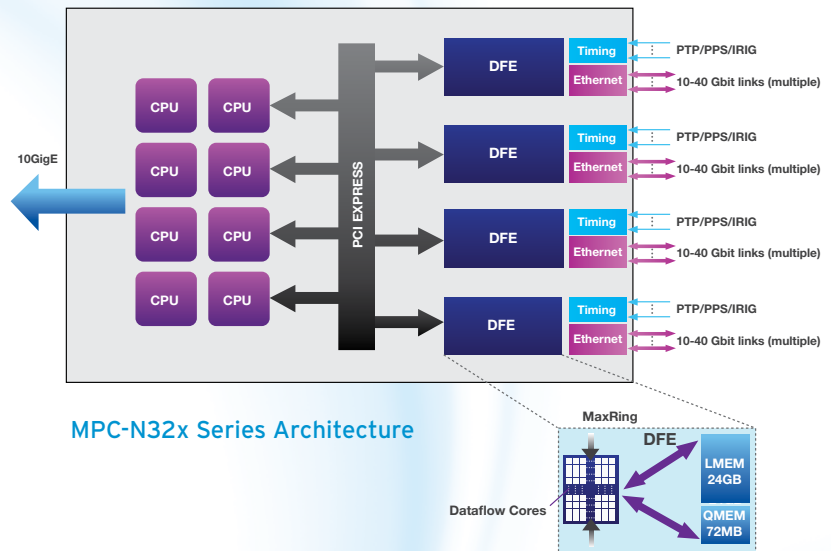
## Endpoint

As network endpoints, MPC-N DFEs implement full network clients or servers. Example applications include low latency financial trading engines, cache or database servers, webservers etc.

A single network port on an MPC-N DFE can be connected to the network, or multiple ports can be utilized to either provide additional bandwidth or to separate logically different parts of a connection (e.g. a trading engine may receive market data on one port, and communicate orders with the exchange on the second port).

# MPC-N Series

MPC-N series nodes provide direct network connections to dataflow engines allowing ultra low latency line-rate processing of multiple data streams. MPC-N32x series nodes use *isca* dataflow engines supporting high performance 10Gbit or 40Gbit connectivity. For MPC-N4x series nodes see page 24.

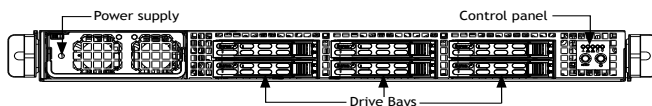


MPC-N32x Series Architecture

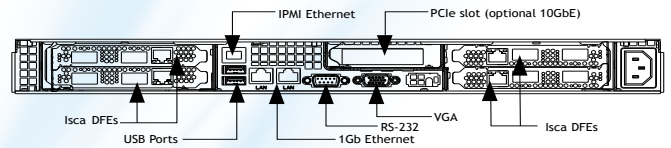
\* See table on page 31 for information on *isca* connectivity options

## MPC-N320 Chassis

Front

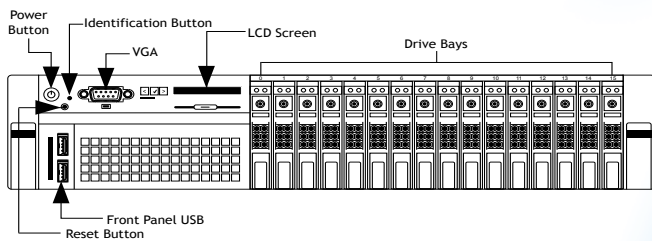


Rear

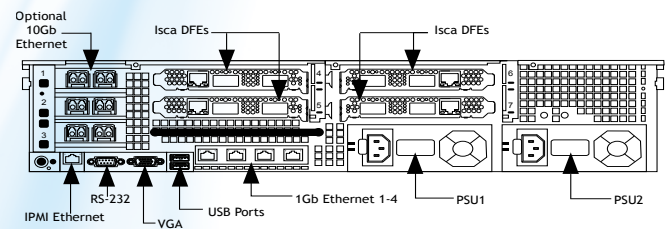


## MPC-N322 Chassis

Front



Rear



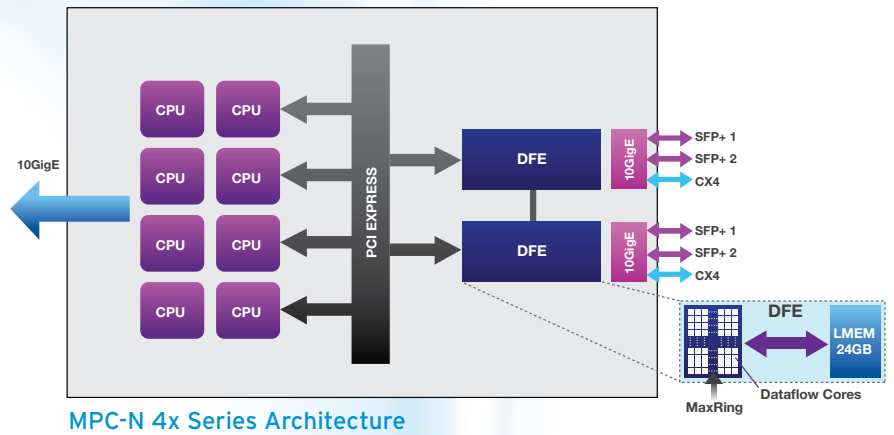


## Specifications

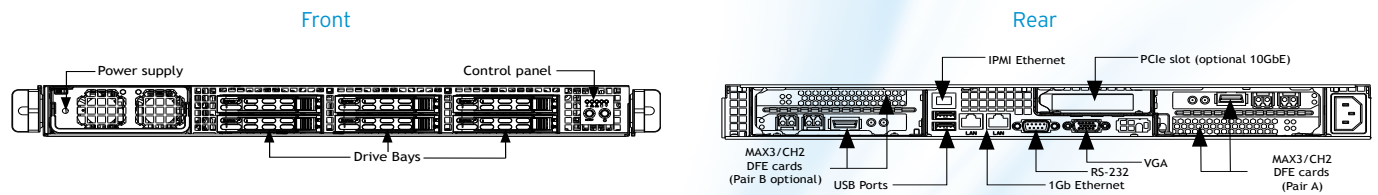
|                          | MPC-N320                                                                                                                                                | MPC-N322                                                                                                                                                                                                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form factor              | 1U                                                                                                                                                      | 2U                                                                                                                                                                                                                                                                        |
| DFE Architecture         | <i>isca</i>                                                                                                                                             |                                                                                                                                                                                                                                                                           |
| DFE LMEM                 | 24 GB per DFE                                                                                                                                           |                                                                                                                                                                                                                                                                           |
| DFE connectivity         | 1, 2 or 3 QSFP 40Gbit connections per DFE (each can be used as 4x10Gbit)                                                                                |                                                                                                                                                                                                                                                                           |
| Number of DFEs           | 1 or 2 or 3 or 4                                                                                                                                        |                                                                                                                                                                                                                                                                           |
| CPUs                     | 2x Intel Xeon X5650<br>12 cores @ 2.66GHz<br>or<br>2x Intel Xeon X5680<br>12 cores @ 3.33GHz                                                            | 2x Intel E5-2640<br>12 cores @ 2.5GHz<br>or<br>2x Intel Xeon E5-2670<br>16 cores @ 2.6GHz<br>or<br>2x Intel Xeon E5-2690<br>16 cores @ 2.9GHz                                                                                                                             |
| Memory                   | 48GB<br>or<br>96GB<br>or<br>192GB                                                                                                                       | 64GB<br>or<br>192GB<br>or<br>384GB                                                                                                                                                                                                                                        |
| Interconnect             | 2x 1GbE and 2x 10GbE<br>or<br>2x 1GbE and 2x 40GbE/FDR Infiniband                                                                                       | 4x 1GbE and 2x 10GbE<br>or<br>4x 1GbE and 2x 40GbE/FDR Infiniband<br>or<br>4x 1GbE and 6x 10GbE                                                                                                                                                                           |
| Storage                  | 2x1TB in RAID1 (1TB usable)<br>or<br>4x1TB in RAID5 (3TB usable)<br>or<br>2x1TB in RAID1 + 2x 240GB SSD in<br>RAID0 (1TB O/S + 480GB scratch<br>usable) | 2x1TB in RAID1 (1TB usable)<br>or<br>2x1TB in RAID1 + 4x 240GB SSD in<br>RAID0 (1TB O/S + 960GB scratch<br>usable)<br>or<br>2x1TB in RAID1 for system<br>plus 14x data disks of:<br>1TB (13TB usable)<br>or<br>600GB 10k RPM (8TB usable)<br>or<br>240GB SSD (3TB usable) |
| RAID                     | Software                                                                                                                                                | Hardware                                                                                                                                                                                                                                                                  |
| Operating System         | CentOS 5/RHEL 5<br>or<br>CentOS 6/RHEL 6                                                                                                                | CentOS 6/RHEL 6                                                                                                                                                                                                                                                           |
| Redundant power supplies |                                                                                                                                                         | ✓                                                                                                                                                                                                                                                                         |
| Power consumption        | 470W (X5680 CPUs, 2 DFEs)                                                                                                                               | 570W (E5-2690 CPUs, 2 DFEs)                                                                                                                                                                                                                                               |
| Weight                   | 16kg                                                                                                                                                    | 25kg (with 2 disks)                                                                                                                                                                                                                                                       |
| Availability             | Now                                                                                                                                                     | Now                                                                                                                                                                                                                                                                       |

# MPC-N4x Series

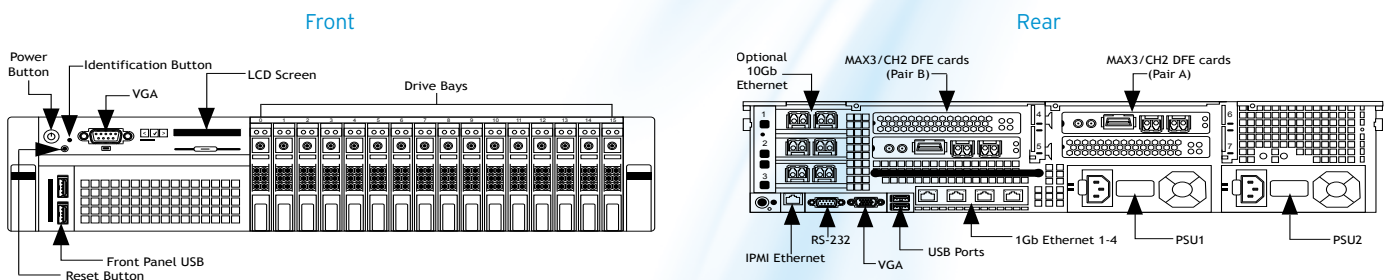
MPC-N4x Series nodes provide direct network connections to *vectis* dataflow engines allowing ultra low latency line-rate processing of multiple 10Gbit data streams.



## MPC-N40 Chassis



## MPC-N42 Chassis



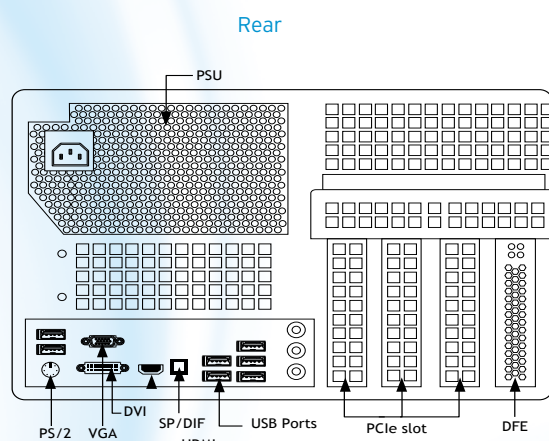
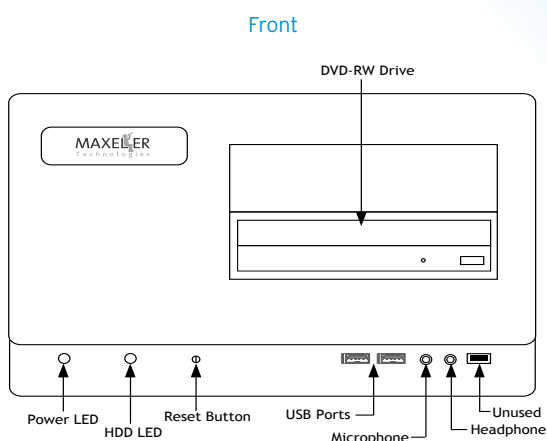
## Specifications

|                          | MPC-N40                                                                                                                                                               | MPC-N42                                                                                                                                                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form factor              | 1U                                                                                                                                                                    | 2U                                                                                                                                                                                                                                                                                                    |
| DfE Architecture         | <i>vectis</i>                                                                                                                                                         |                                                                                                                                                                                                                                                                                                       |
| DfE LMEM                 | 24 GB per DfE                                                                                                                                                         |                                                                                                                                                                                                                                                                                                       |
| DfE connectivity         | 2x SFP+ and 1x CX4 10Gbit connections per DfE                                                                                                                         |                                                                                                                                                                                                                                                                                                       |
| Number of DfEs           | 1 or 2                                                                                                                                                                |                                                                                                                                                                                                                                                                                                       |
| CPUs                     | 2x Intel Xeon X5650<br>12 cores @ 2.66GHz<br><b>or</b><br>2x Intel Xeon X5680<br>12 cores @ 3.33GHz                                                                   | 2x Intel E5-2640<br>12 cores @ 2.5GHz<br><b>or</b><br>2x Intel Xeon E5-2670<br>16 cores @ 2.6GHz<br><b>or</b><br>2x Intel Xeon E5-2690<br>16 cores @ 2.9GHz                                                                                                                                           |
| Memory                   | 48GB<br><b>or</b><br>96GB<br><b>or</b><br>192GB                                                                                                                       | 64GB<br><b>or</b><br>192GB<br><b>or</b><br>384GB                                                                                                                                                                                                                                                      |
| Interconnect             | 2x 1GbE and 2x 10GbE<br><b>or</b><br>2x 1GbE and 1x FDR Infiniband                                                                                                    | 4x 1GbE and 2x 10GbE<br><b>or</b><br>4x 1GbE and 2x 10GbE and<br>1x FDR Infiniband<br><b>or</b><br>4x 1GbE and 6x 10GbE                                                                                                                                                                               |
| Storage                  | 2x1TB in RAID1 (1TB usable)<br><b>or</b><br>4x1TB in RAID5 (3TB usable)<br><b>or</b><br>2x1TB in RAID1 + 2x 240GB SSD in<br>RAID0 (1TB O/S + 480GB scratch<br>usable) | 2x1TB in RAID1 (1TB usable)<br><b>or</b><br>2x1TB in RAID1 + 4x 240GB SSD in<br>RAID0 (1TB O/S + 960GB scratch<br>usable)<br><b>or</b><br>2x1TB in RAID1 for system<br>plus 14x data disks of:<br>1TB (13TB usable)<br><b>or</b><br>600GB 10k RPM (8TB usable)<br><b>or</b><br>240GB SSD (3TB usable) |
| RAID                     | Software                                                                                                                                                              | Hardware                                                                                                                                                                                                                                                                                              |
| Operating System         | CentOS 5/RHEL 5<br><b>or</b><br>CentOS 6/RHEL 6                                                                                                                       | CentOS 6/RHEL 6                                                                                                                                                                                                                                                                                       |
| Redundant power supplies |                                                                                                                                                                       | ✓                                                                                                                                                                                                                                                                                                     |
| Power consumption        | 450W (X5680 CPUs)                                                                                                                                                     | 550W (E5-2690 CPUs)                                                                                                                                                                                                                                                                                   |
| Weight                   | 16kg                                                                                                                                                                  | 25kg (with 2 disks)                                                                                                                                                                                                                                                                                   |
| Availability             | Now                                                                                                                                                                   | Now                                                                                                                                                                                                                                                                                                   |

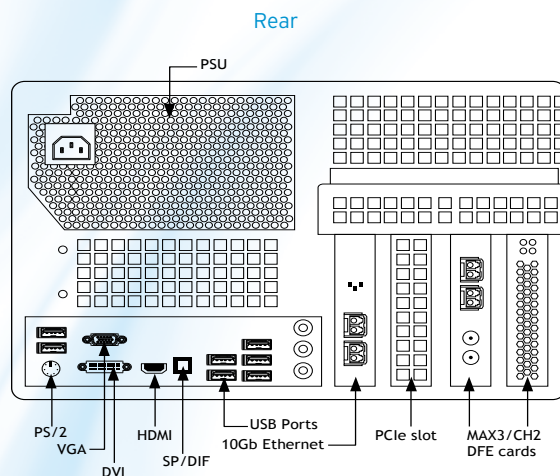
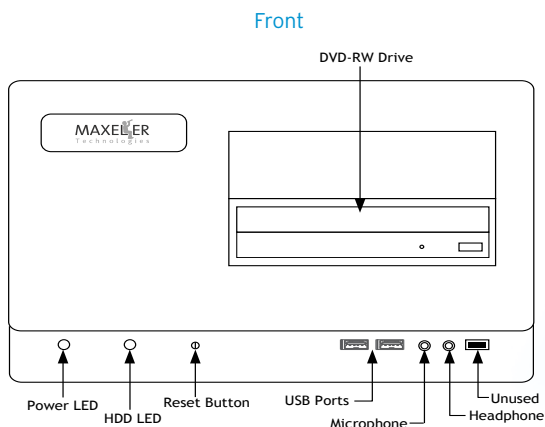
# Desktop

MaxWorkstation and MaxWorkstation10G are small form factor PCs that bring the power of dataflow computing to the desktop. Ideal for software developers as well as users who desire a dedicated compute resource on their desk.

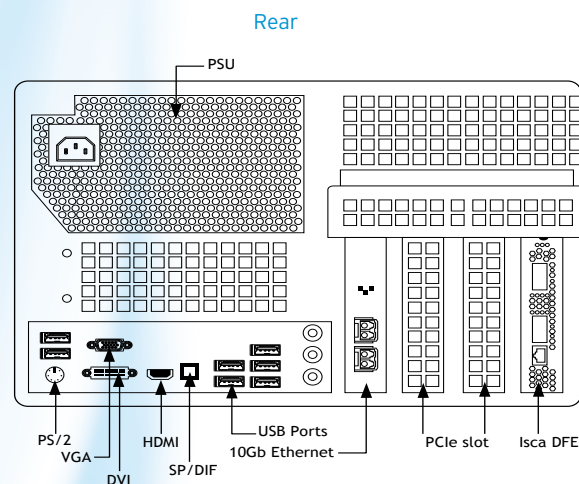
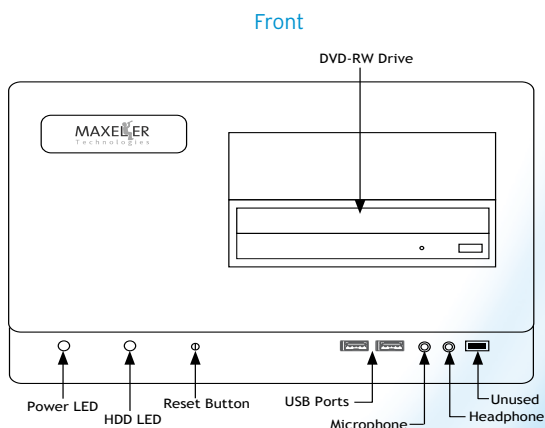
## MaxWorkstation Chassis



## MaxWorkstation10G Chassis



## MaxWorkstation40G Chassis





## Specifications

|                          | MaxWorkstation                                                  | MaxWorkstation10G                             | MaxWorkstation40G                                                                |
|--------------------------|-----------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------|
| Form factor              | Small form factor Desktop                                       |                                               |                                                                                  |
| DFE Architecture         | <i>vectis</i> <b>or</b> <i>coria</i> <b>or</b> <i>maia</i>      | <i>vectis</i>                                 | <i>isca</i>                                                                      |
| DFE DRAM                 | 24GB (vectis) <b>or</b> 48GB (coria <b>or</b> maia)             | 24GB                                          | 24GB                                                                             |
| DFE connectivity         | None                                                            | 2x SFP+ and 1x CX4 10Gbit connections per DFE | 1, 2 or 3 QSFP 40Gbit connections (each can be used as 4x10Gbit)                 |
| Number of DFEs           | 1 <b>or</b> 2                                                   | 1                                             | 1                                                                                |
| CPUs                     | Intel core i7-2600S<br>4 cores at 2.8GHz                        |                                               |                                                                                  |
| Memory                   | 32GB                                                            |                                               |                                                                                  |
| Interconnect             | 2x 1GbE<br><b>or</b><br>2x 1GbE and 2x 10GbE (with 1 DFE only)  | 2x 1GbE<br><b>or</b><br>2x 1GbE and 2x 10GbE  | 2x 1GbE<br><b>or</b><br>2x 1GbE and 2x10GbE<br><b>or</b><br>2x 1GbE and 2x 40GbE |
| Storage                  | 2x500GB in RAID1<br>(500GB usable)                              |                                               |                                                                                  |
| Operating System         | CentOS 6                                                        | CentOS 6                                      | CentOS 6                                                                         |
| Redundant power supplies |                                                                 |                                               |                                                                                  |
| Power consumption        | 140W (1 DFE)                                                    | 150W                                          | 160W                                                                             |
| Weight                   | 8kg                                                             | 8kg                                           | 8kg                                                                              |
| Availability             | Now ( <i>vectis</i> + <i>maia</i> )<br>Q4 2013 ( <i>coria</i> ) | Now                                           | Now                                                                              |

# Development Tools

Maxeler software tools enable scientists and software engineers to rapidly develop and deploy accelerated applications on our high-performance dataflow systems. Maxeler development tools are supported on Red Hat / CentOS versions 5 and 6.

MaxCompiler is a general-purpose development tool suite that is required for all DFE development. Maxeler's Domain Specific Language (DSL) libraries work in conjunction with MaxCompiler to enable rapid application development in particular problem domains. We recommend selecting the right mix of development tools depending on your target application area.

**License terms:** Maxeler development tools are licensed per developer seat (node locked), with annual maintenance subscription. Maintenance subscription entitles you to receive updated versions of the software released during the maintenance period.

|                     | MaxCompiler | Domain Specific Language Modules |                |
|---------------------|-------------|----------------------------------|----------------|
|                     |             | MaxGenFD                         | MaxCompilerMPT |
| Financial analytics | ✓           |                                  |                |
| Low latency trading | ✓           |                                  | ✓              |
| Networking          | ✓           |                                  | ✓              |
| Geoscience          | ✓           | ✓                                |                |
| Life science        | ✓           |                                  |                |
| General HPC         | ✓           |                                  |                |

## MaxCompiler

MaxCompiler includes the Maxeler Kernel compiler, Manager compiler, MaxIDE graphical development environment and high speed DFE simulator. Select the edition that matches your target DFE architecture.

|            | S Edition | V Edition | A Edition | C Edition |
|------------|-----------|-----------|-----------|-----------|
| Simulation | ✓         | ✓         | ✓         | ✓         |
| Vectis     |           | ✓         |           | ✓         |
| Coria      |           |           | ✓         | ✓         |
| Maia       |           |           | ✓         | ✓         |
| Isca       |           |           | ✓         | ✓         |

## Domain Specific Languages

### MaxGenFD

MaxGenFD enables geoscientists to rapidly develop 3D finite difference applications such as seismic Forward Modeling, Reverse Time Migration and Full Waveform Inversion. It includes support for automatic parallelization across multiple DFEs, wavefield and earth model compression and in-built boundary conditions.

### MaxCompilerMPT

MaxCompilerMPT enables development of low-latency and/or high throughput networking applications. It includes UDP and TCP/IP protocol support, network frame kernel compiler, latency optimized compilation flow.

# Securities Trading Interfaces

Maxeler provide a suite of fully maintained Market Data and Order Entry Interfaces to financial venues. Each Interface has a complete hardware and software configuration interface to manage all aspects of exchange connectivity.

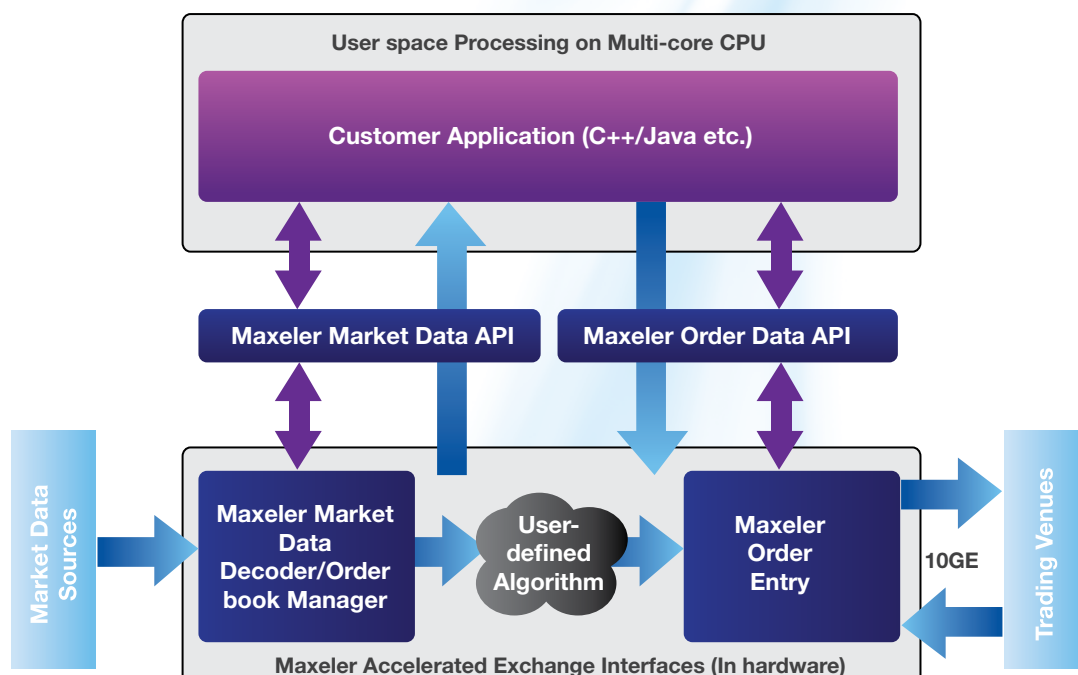
## Market Data Interfaces

- Real-time A/B Feed arbitration and gap-fill
- Line rate operation
- Customizable order book and trade view
- Configurable mid-message updates

## Order Entry Interfaces

- Multi-session support with Hardware / Software interlocking for volume / order control
- Customizable order submission through ultra-low latency hardware interface
- Low-latency full-featured Software Interface
  - New Order / Replace / Cancel
- Session and User Management
  - Gap Fill / Replay
  - Heartbeats

|                                        | Market Data    | Order Entry    |
|----------------------------------------|----------------|----------------|
| CME Globex FIX/FAST / iLink FIX        | Production     | Production     |
| Eurex NTA ETI / EMDI                   | Production     | Production     |
| ICE iMPact Multicast / FIX             | Planned (2013) | Planned (2013) |
| BATS Multicast PITCH/BZX/BYX BOE       | Planned (2013) | Planned (2013) |
| NASDAQ TotalView-ITCH 4.1 / OUCH 4.2   | Planned (2013) | Planned (2013) |
| DirectEdge EdgeBook Multicast MD / MEP | Planned (2013) | Planned (2013) |
| NYSE Euronext                          | Planned (2013) | Planned (2013) |



# Dataflow Engines

Maxeler hardware systems contain multiple Dataflow Engines (DFEs) providing the dataflow computing resource. Each DFE contains a large configurable logic fabric, off-chip memory and interconnect. DFEs have several kinds of memory resources:

- FMEM: ultra fast on-chip memory
- QMEM: fast off-chip memory with low latency
- LMEM: large off-chip memory for bulk data storage

DFEs also have interconnect to/from CPU (via Infiniband in MPC-X Series systems, or PCI Express in MPC-C or MPC-N Series) and MaxRing connections to other DFEs. Some DFEs also support direct low-latency network connectivity to the DFE.

Maxeler provides several different DFE architectures optimized for different use cases. Raw specifications are tabulated below.

|                             | Vectis                                    | Coria                                      | Maia                                         | Isca                                                        |
|-----------------------------|-------------------------------------------|--------------------------------------------|----------------------------------------------|-------------------------------------------------------------|
| Compute resources           |                                           |                                            |                                              |                                                             |
| Logic fabric size           | 475k                                      | 460k                                       | 700k                                         | 950k                                                        |
| Multipliers                 | 2000                                      | 3000                                       | 4000                                         | 700                                                         |
| On-chip FMEM                |                                           |                                            |                                              |                                                             |
| Capacity                    | 4MB                                       | 5MB                                        | 6MB                                          | 6.5MB                                                       |
| Max bandwidth               | 10.4TB/s                                  | 11.0TB/s                                   | 14.0TB/s                                     | 14.4TB/s                                                    |
| Average latency             | 1 cycle e.g. 5ns at 200MHz                |                                            |                                              |                                                             |
| Off-chip LMEM               |                                           |                                            |                                              |                                                             |
| Capacity                    | 24GB                                      | 48GB                                       | 48GB                                         | 24GB                                                        |
| Max bandwidth               | 38.4GB/s                                  | 51.2GB/s                                   | 76.8GB/s                                     | 38.4GB/s                                                    |
| Average latency             | 250ns                                     |                                            |                                              |                                                             |
| Off-chip QMEM               |                                           |                                            |                                              |                                                             |
| Capacity                    | –                                         | –                                          | –                                            | 72MB                                                        |
| Max bandwidth               | –                                         | –                                          | –                                            | 9.9GB/s<br>bidirectional (19.8<br>GB/s aggregate)           |
| Average latency             | –                                         | –                                          | –                                            | 32ns                                                        |
| Interconnect                |                                           |                                            |                                              |                                                             |
| Max bandwidth to CPU        | 2GB/s                                     | 3GB/s                                      | 3GB/s                                        | 3GB/s                                                       |
| Max MaxRing bandwidth*      | 2GB/s                                     | 1GB/s (standard**) 4GB/s (5gbit/s optical) | 1GB/s (standard**) 4.7GB/s (5gbit/s optical) | 4.7GB/s (5gbit/s optical)                                   |
| Direct network connectivity | 2x SFP+ & 1x CX4 10Gbit (via CH2 adapter) | –                                          | –                                            | 1, 2 or 3 QSFP 40Gbit connections (can be used as 4x10Gbit) |

\* MaxRing Bandwidth is bi-directional for each link direction. So aggregate MaxRing bandwidth per DFE is 4x this number.

\*\* Only available in MPC-X



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## Isca Port Configurations

Isca DFEs are available in three different port configurations, with different connectivity supported in each configuration.

|                                                  | A                    | B       | C        |
|--------------------------------------------------|----------------------|---------|----------|
|                                                  | Network Connectivity |         |          |
| User-configurable QSFP ports (4x10Gbps/1x40Gbps) | 1                    | 2       | 3        |
| Full-duplex Bandwidth                            | 40 Gbps              | 80 Gbps | 120 Gbps |
|                                                  | Precision Timing     |         |          |
| PTPv2 (IEEE1588-2008)                            | ✓                    | ✓       | ✓        |
| PPS (pulse-per-second)                           | ✓                    |         |          |
| IRIG-B                                           | ✓                    |         |          |

Each user-configurable QSFP port can be configured as a single 40Gbit link or up to four 10Gbit links. The ports can use QSFP to QSFP, QSFP to 4xSFP+ direct attach cables or 40Gbps / 4x10Gbps QSFP optical modules. For single SFP+ connectivity, QSFP to SFP+ adapters are available.

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## Next Steps

To request more information or a quote, please contact your Maxeler representative, email us at [info@maxeler.com](mailto:info@maxeler.com) or contact us by telephone:

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